PIC 10A 1A

TA: Bumsu Kim

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Today…

- HW2 Review
- Exercises on Loops
  - Fibonacci/Prime Number Generator
- Exercise on Random Numbers
  - Random Walk Simulator
- HW3 Hints

HW2 Review

HW2 Review

- Exercise 1: Consider three `if` clauses (you don't need `else`. why?)

```
// Exercise 1: Leap year
int year = 1582; // initialization (number doesn't matter here, but 1582 makes sense)
cout << "Enter year strictly after 1581 AD: ";
cin >> year;
cout << "This is : ";
bool leap_yr = false;
if(year % 4 == 0) {
        leap_yr = true;
}
if (year % 100 == 0) {
        leap_yr = false;
}
if (year % 400 == 0) {
        leap_yr = true;
}
if (leap_yr) {
        cout << "A leap year\n";
}
else {
        cout << "Not a leap year\n";
}
```

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HW2 Hints

- Exercise 2 (Solution 1: Strategy B from the slides in Week 4)

```
// Exercise 2: Num2Rome
int n = 1;
cout << "Enter integer
   from 1 to 3,999: ";
cin >> n;
cout << "Roman numeral: ";
while (n >= 1000) {
     cout << "M";
     n -= 1000;
}
```

```
// now n<1000
if (n >= 900) {
     cout << "CM";
     n -= 900;
}
if (n >= 500) {
     cout << "D";
     n -= 500;
}
if (n >= 400) {
     cout << "CD";
     n -= 400;
}
while (n >= 100) {
     cout << "C";
     n -= 100;
}
```

```
// now n<100
if (n >= 90) {
     cout << "XC";
     n -= 90;
}
if (n >= 50) {
     cout << "L";
     n -= 50;
}
if (n >= 40) {
     cout << "XL";
     n -= 40;
}
while (n >= 10) {
     cout << "X";
     n -= 10;
}
```

```
// now n<10
if (n >= 9) {
     cout << "IX";
     n -= 9;
}
if (n >= 5) {
     cout << "V";
     n -= 5;
}
if (n >= 4) {
     cout << "IV";
     n -= 4;
}
while (n >= 1) {
     cout << "I";
     n -= 1;
}
cout << endl;
```

Simple idea, but quite long

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HW2 Hints

- Exercise 2 (Solution 2: Strategy A from the slides in Week 4)

```
string romans = "IVXLCDM", output = "";
int place = 0;
while (n > 0) {
    if (n % 10 > 0){
        if (n % 10 == 9) { // "IX", "XC", or "CM" (02, 24, 46)
            output = romans[place*2] + (romans[place*2 + 2] + output);
            n -= 9;
        }
        else if (n % 10 == 4) { // "IV", "XL", "CD" (01, 23, 45)
            output = romans[place * 2] + (romans[place * 2 + 1] + output);
            n -= 4;
        }
        while (n % 5 > 0) { // #: 1,2,3, 6,7,8 (append I, X, C or M)
            output = romans[place * 2] + output;
            --n;
        }
        if (n % 10 == 5) { // V, L, or D
            output = romans[place * 2 + 1] + output;
            n -= 5;
        }
    }
    n /= 10;
    ++place;
}
cout << output << endl;
```

More Challenging Loop Exercises

Fibonacci/Prime Number Generator

Why are we doing this?

- There was a feedback from one of you:

> Topics you want to see in Discussion 1A *
>
> Be as specific as you can. For instance, Visual Studio tips and shortcuts, More examples on C++ arrays, HW 2 hints on some specific functionality, etc.
>
> Questions similar to those in homework and exams. p.s. I am confused about those challenging questions and I wonder how helpful they are for later programming

- Thanks for the feedback! That's a great question: how helpful are they?
- Those "challenging exercises" are in fact from HW/Quiz/Exam problems in the previous PIC10A courses
  - As you can see in the acknowledgement in the lecture slides, some instructors share their slides and HW/Exam problems

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Fibonacci/Prime Number Generator

- 1. Fibonacci Number Generator
  - The user inputs a positive integer (**n**)
  - The program outputs the **n**-th Fibonacci number F_n
  - Fibonacci numbers satisfy: $F_{n+2} = F_{n+1} + F_n$
    (c.f. rabbit population puzzle posed by Fibonacci)

- 2. Prime Number Generator
  - The user inputs a positive integer (**n**)
  - The program outputs the **n**-th prime number P_n
  - First few prime numbers are: 2, 3, 5, 7, 11, 13, …

- Input and Output format:

```
Please enter a positive integer.
5
The 5th Fibonacci number is 5.
The 5th prime number is 11.
```

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Hints

- Fibonacci numbers are relatively simple
  - Just use a for loop
  - Can be done using 3 variables

- Prime numbers is more difficult
  - Break the problem into smaller pieces
  - First create a variable `int cnt` that counts the number of primes so far
  - You can then use a while loop

```
while(cnt < n) { // we want the n-th prime…
      // increment the current number
      // check if the current number is prime
      // if so, increment cnt
      // if not, don't increment cnt
```

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Hints

- Primality test (smaller piece of the entire problem)
- An integer is not prime if it has a divisor greater than 1
- So, consider using a loop to check if it has a divisor
- *Mathematical fact: you only need to check if it has a divisor $\leq \sqrt{N}$ → how to implement this w/o using `std::sqrt()`?
- You can set a flag `bool isPrime` to store the primality test result

```
for(div = 2; div*div <= num; ++div ) {
      // check if div divides num
      // if so, set isPrime false and exit from this (inner) loop
      // if not, do nothing
```

Q: what should be the *default value* for `isPrime`?

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Exercise on Random Numbers

Random Walk Simulator

Summary for Random Numbers in C++

- `#include <cstdlib>` for `rand()` and `srand()`

- `#include <ctime>` for `time()`

- Use `time(nullptr)` to generate a seed:
  - `srand(time(nullptr));`
  - And then use `rand()`

- Intervals:
  - `r_double = (rand()*1./RAND_MAX)*(B-A) + A;`
  - `r_int = A + rand()%(B-A+1);`

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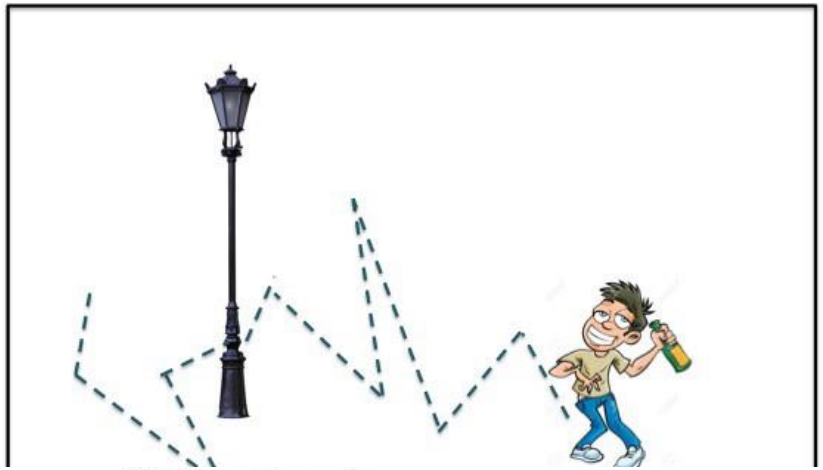
Exercise – Random Walk Simulator

(will be revisited on Thursday)

- Suppose someone walks in a constant speed, but chooses the direction randomly in each step
- The path (to be more precise, this *stochastic process*) is called a random walk

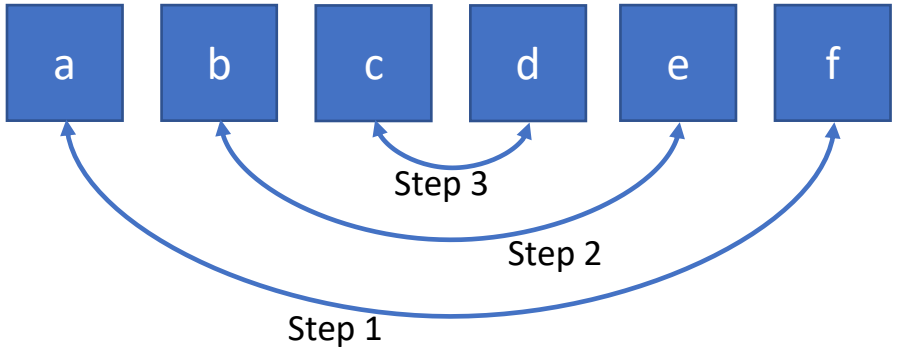


- We will simulate a 1-D random walk:
  - An object can only move forward or backward at each step
  - When the object hits the wall (upper/lower bounds), it stops walking
  - The time that it hits the wall is called the "stopping time"
  - Important in mathematical finance

HW3 Hints

HW3 Hints

- Exercise 1:
  - Consider “swapping” two characters in the string



Original string:	abcdef
after Step 1:	fbcdea
after Step 2:	fecdba
after Step 3:	fedcba

Then stop!

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HW3 Hints

- Exercise 2: Finding the 2nd smallest number in the given stream
  - Recall Grade Calculator ver.2
  - How did we find the smallest number there?
  - It should be very similar
    - Consider *two* variables instead of *one* (because we need the second smallest, and to do this you *also* need to keep track of the smallest)
    - And, add some more if's for them

  - Don't forget the same numbers don't count:
    - If the input were: "1 2 1 8 8 9 Q", the smallest and the second smallest are:
    - 1 → 1, (undecided)
    - 2 → 1, 2
    - 1 → 1, 2 (without a care, it may become "1, 1")
    - 8 → 1, 2
    - 9 → 1, 2
    - Q → quit, and output 2

Note: if the user inputs only *one* number, you should set
the second smallest = that number
(See the last test case in the instruction)

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Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)

Click this link

At this point, you may want to request
“more hw hints”,
“more challenging examples/exercises”,
“more details on the classroom examples”, etc.
Please let me know what you want!